

16-Bit RISC Processor Design Using Verilog HDL

Major Project Report

Submitted By

Aparna Dubey and Aayush Jhanwar

Introduction

What is a 16-bit RISC Processor?

A Reduced Instruction Set Computer (RISC) processor is a processor architecture that executes a small set of simple instructions efficiently. A 16-bit RISC processor processes 16-bit data and performs operations such as arithmetic, logical, memory access, and instruction execution. It consists of several modules working together to fetch, decode, execute, and store data.

Importance of RISC Processor

- Simple and efficient processor architecture.
- Faster execution using simple instructions.
- Lower hardware complexity.
- Easy to understand and implement.
- Widely used in embedded systems and microcontrollers.

This project presents the design and verification of a **16-Bit RISC Processor** using **Verilog HDL**. The processor integrates multiple hardware modules including the Program Counter, Instruction Memory, Register File, Control Unit, Arithmetic Logic Unit (ALU), Data Memory, and CPU Top module. Functional verification was carried out using **Xilinx Vivado 2025.2** through behavioral simulation and RTL elaboration.

Objective

The objectives of this project are:

- Design a simplified 16-bit RISC Processor using Verilog HDL.
- Implement the Program Counter.

- Design the Instruction Memory.
- Design the Register File.
- Implement the Arithmetic Logic Unit (ALU).
- Design the Control Unit.
- Implement the Data Memory.
- Integrate all modules into a CPU Top module.
- Verify functionality using behavioral simulation.
- Generate the RTL schematic.

Tools Used

- Verilog HDL
- Xilinx Vivado 2025.2
- Visual Studio Code
- GitHub
- Windows 11

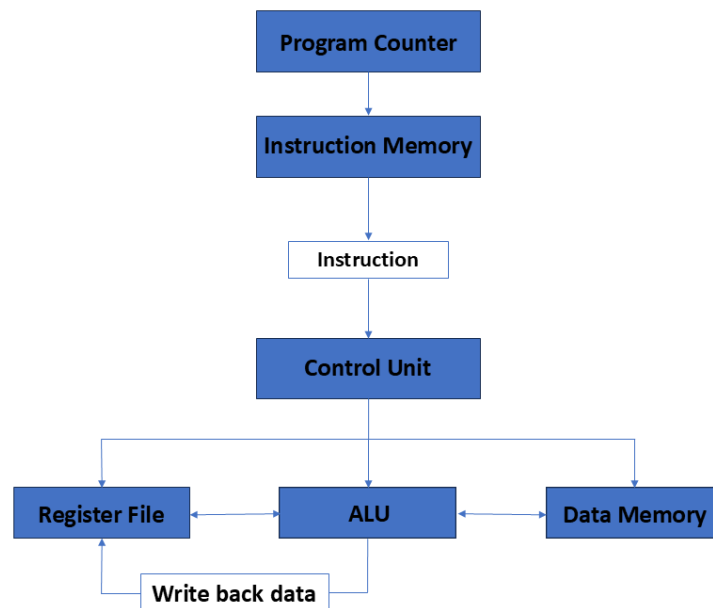
Block Diagram

The processor consists of:

- Program Counter (PC)
- Instruction Memory
- Control Unit
- Register File
- Arithmetic Logic Unit (ALU)
- Data Memory
- Multiplexer (MUX)
- CPU Top Module

The Program Counter generates instruction addresses. The Instruction Memory stores instructions. The Control Unit decodes instructions and generates

control signals. The Register File stores operands. The ALU performs arithmetic and logical operations. The Data Memory is used for load/store operations. All modules are integrated in the CPU Top module.

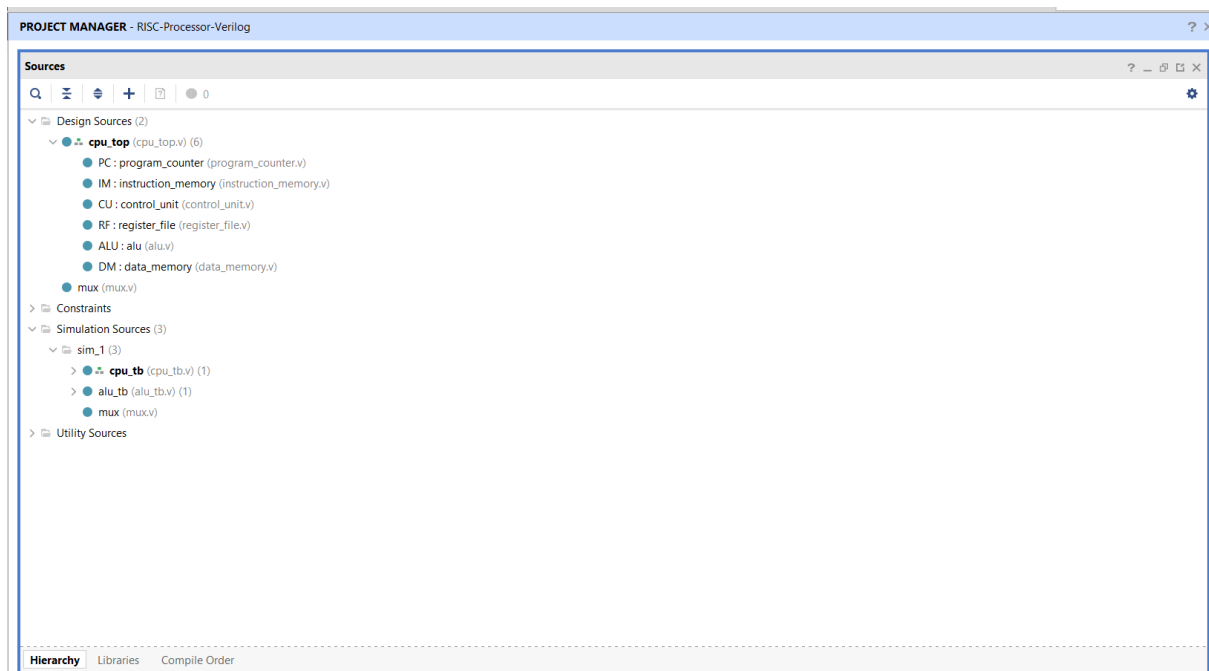


Verilog Modules

Modules Used:

- program_counter.v
- instruction_memory.v
- register_file.v
- alu.v
- control_unit.v
- data_memory.v
- mux.v
- cpu_top.v

The **CPU Top** module integrates all submodules to perform instruction execution.



Testbench

A testbench was developed to verify the processor functionality.

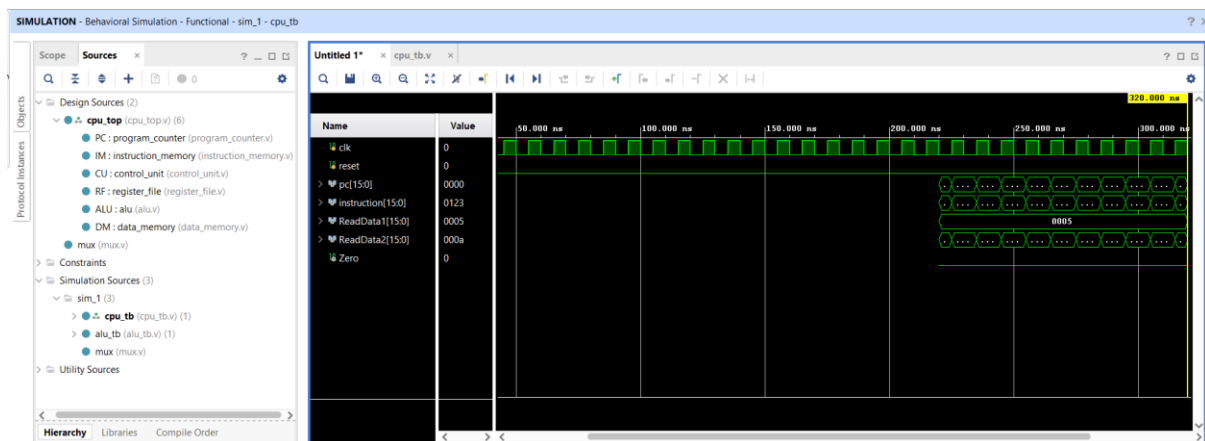
Test Cases

Test No.	Operation
1	Program Counter Increment
2	Instruction Fetch
3	Register Read
4	Register Write
5	ADD Operation
6	SUB Operation
7	AND Operation
8	OR Operation
9	XOR Operation

Test No.	Operation
10	Memory Read/Write

Waveforms

The processor was simulated using **Xilinx Vivado 2025.2**. The simulation verified the correct operation of the Program Counter, Instruction Memory, Register File, ALU, Control Unit, and Data Memory. The waveform showed successful instruction fetching, register operations, ALU execution, and memory access.

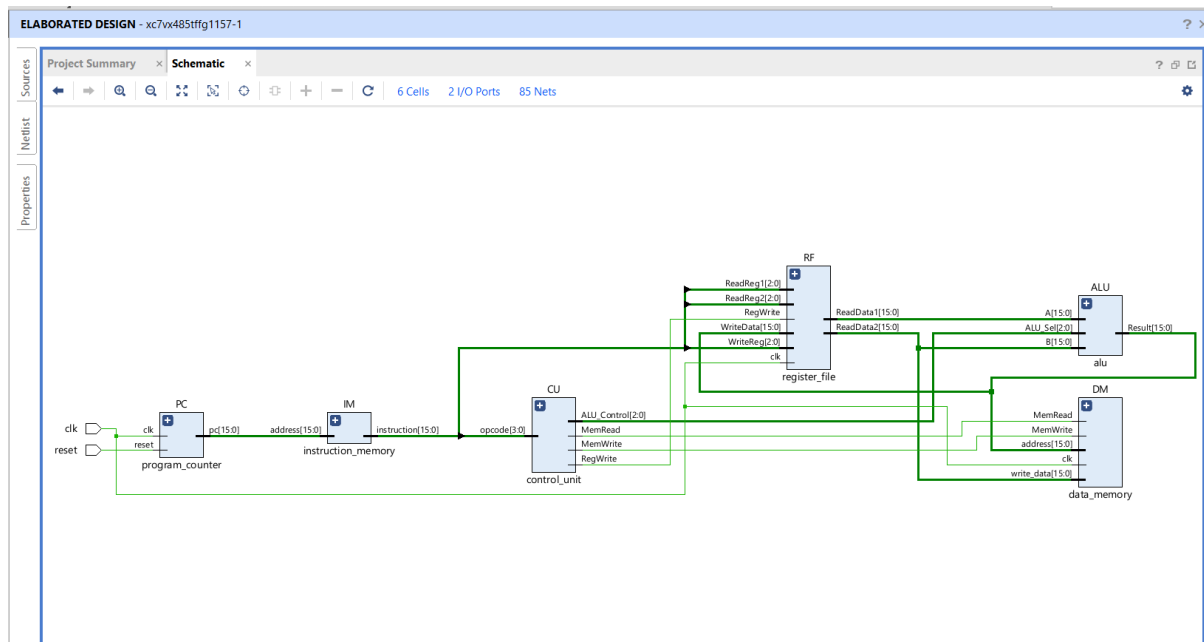


Observations

- Program Counter incremented correctly.
- Instructions were fetched successfully from Instruction Memory.
- Register File performed read and write operations correctly.
- ALU executed arithmetic and logical operations successfully.
- Control Unit generated appropriate control signals.
- Data Memory responded correctly to memory operations.
- CPU Top module successfully integrated all hardware modules.
- Behavioral simulation produced expected outputs.

RTL Schematic

The RTL schematic represents the hardware structure generated from the Verilog code before synthesis. It shows the interconnection between the Program Counter, Instruction Memory, Register File, Control Unit, ALU, and Data Memory.



Conclusion

The **16-Bit RISC Processor** was successfully designed and verified using **Verilog HDL**. Individual hardware modules were implemented and integrated into a complete processor architecture. Behavioral simulation and RTL elaboration in **Xilinx Vivado 2025.2** confirmed the correct functionality of the processor. The project provides a practical understanding of processor architecture and digital system design and serves as a foundation for developing more advanced processors in the future.